

СОДЕРЖАНИЕ

ЗАДАНИЕ 1	2
ЗАДАНИЕ 2	23
ЗАДАНИЕ 3	29
ЗАДАНИЕ 4	39
ЗАДАНИЕ 5	42
ЗАДАНИЕ 6	48
ЗАДАНИЕ 7	52
ЗАДАНИЕ 8	61

ЗАДАНИЕ 1

1) Для заданной на схеме schema-lab5 сети, состоящей из управляемых коммутаторов, маршрутизаторов и персональных компьютеров

выполнить планирование и документирование адресного пространства и назначить статические адреса всем устройствам.

nb! Каждое соединение маршрутизатора с маршрутизатором - это отдельная сеть.

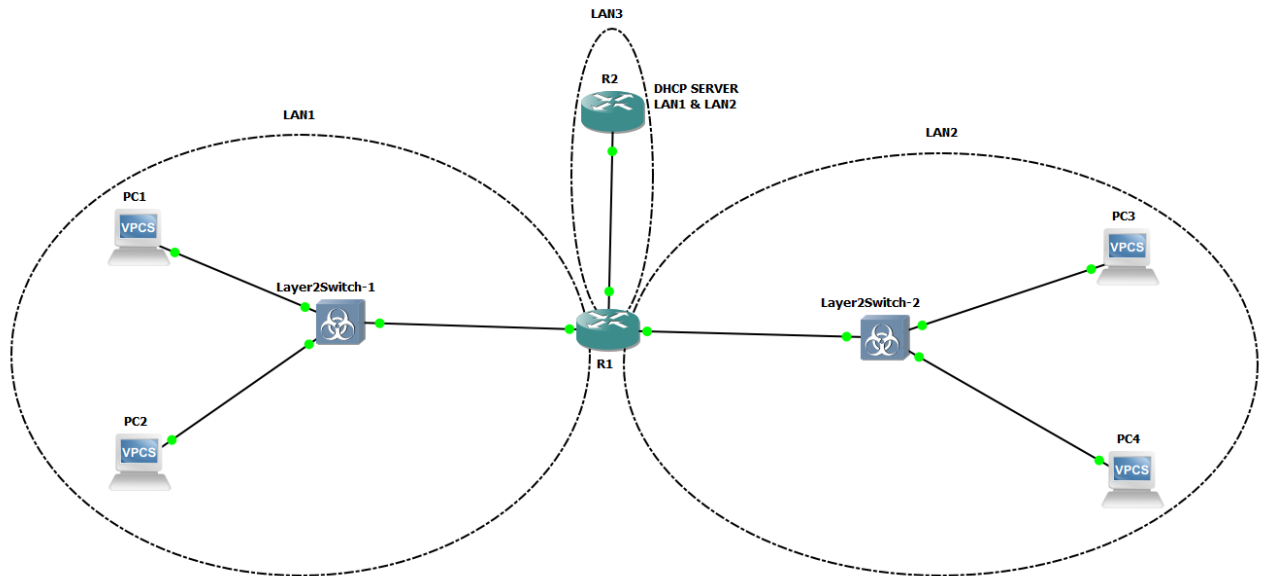


Рисунок 1 — Топология сети для настройки протокола DHCP

R1#enable

R1#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	administratively down	down
FastEthernet1/0	unassigned	YES	unset	administratively down	down

R1#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge

S - Switch, H - Host, I - IGMP, r - Repeater

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
-----------	---------------	---------	------------	----------	---------

R1#

R2#enable

R2#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.123.2	YES	NVRAM	up	up
FastEthernet1/0	192.168.26.2	YES	NVRAM	up	up

R2#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
vIOS-L2-01	Fas 0/0	135	R S I	IOSv	Gig 0/1

R2#

R3#enable

R3#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.123.3	YES	NVRAM	up	up
FastEthernet1/0	192.168.37.3	YES	NVRAM	up	up
FastEthernet2/0	192.168.38.3	YES	NVRAM	up	up

R3#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
vIOS-L2-01	Fas 0/0	147	R S I	IOSv	Gig 0/2

R7	Fas 1/0	125	R S I	3640	Fas 0/0
----	---------	-----	-------	------	---------

R8	Fas 2/0	132	R S I	3640	Fas 0/0
----	---------	-----	-------	------	---------

R3#

R4#enable

R4#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Prot ocol
-----------	------------	-----	--------	--------	-----------

FastEthernet0/0	192.168.1.1	YES NVRAM	up	up
FastEthernet1/0	192.168.45.4	YES NVRAM	up	up
FastEthernet2/0	unassigned	YES NVRAM	administratively down	down

R4#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
R5	Fas 1/0	152	R S I	3640	Fas 1/0

R4#

R5#enable

R5#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.2.1	YES	NVRAM	up	up
FastEthernet1/0	192.168.45.5	YES	NVRAM	up	up
FastEthernet2/0	192.168.51.5	YES	NVRAM	up	up

R5#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
R4	Fas 1/0	159	R S I	3640	Fas 1/0

R5#

R6#enable

R6#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Prot	ocol
FastEthernet0/0	unassigned	YES	unset	administratively down	down	down
FastEthernet1/0	unassigned	YES	unset	administratively down	down	down
FastEthernet2/0	unassigned	YES	unset	administratively down	down	down

R6#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
-----------	---------------	---------	------------	----------	---------

R6#

R7#'enable

% Unknown command or computer name, or unable to find computer address

R7#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.37.7	YES	NVRAM	up	up
FastEthernet1/0	192.168.4.1	YES	NVRAM	up	up
FastEthernet2/0	192.168.67.7	YES	NVRAM	up	up

R7#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
R3	Fas 0/0	167	R S I	3640	Fas 1/0

R7#

R8#enable

R8#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.38.8	YES	NVRAM	up	up
FastEthernet1/0	192.168.5.1	YES	NVRAM	up	up

R8#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
-----------	---------------	---------	------------	----------	---------

R3 Fas 0/0 150 R S I 3640 Fas 2/0
R8#

PC1> ip 172.16.1.10/24 172.16.1.1

Checking for duplicate address...

PC1 : 172.16.1.10 255.255.255.0 gateway 172.16.1.1

PC1>

PC1> save

Saving startup configuration to startup.vpc

. done

PC1> show ip

NAME : PC1[1]

IP/MASK : 172.16.1.10/24

GATEWAY : 172.16.1.1

DNS :

MAC : 00:50:79:66:68:00

LPORT : 27304

RHOST:PORT : 127.0.0.1:27305

MTU : 1500

PC1>

PC2> ip 172.16.2.10/24 172.16.2.1

Checking for duplicate address...

PC2 : 172.16.2.10 255.255.255.0 gateway 172.16.2.1

PC2> save

Saving startup configuration to startup.vpc

. done

PC2> show ip

NAME : PC2[1]
IP/MASK : 172.16.2.10/24
GATEWAY : 172.16.2.1
DNS :
MAC : 00:50:79:66:68:01
LPORT : 27306
RHOST:PORT : 127.0.0.1:27307
MTU : 1500

PC3> ip 10.2.3.10/24 10.2.3.1

Checking for duplicate address...

PC3 : 10.2.3.10 255.255.255.0 gateway 10.2.3.1

PC3> save

Saving startup configuration to startup.vpc

. done

PC3> show ip

NAME : PC3[1]
IP/MASK : 10.2.3.10/24
GATEWAY : 10.2.3.1
DNS :
MAC : 00:50:79:66:68:02
LPORT : 27308
RHOST:PORT : 127.0.0.1:27309
MTU : 1500

PC4> ip 10.2.4.10/24 10.2.4.1

Checking for duplicate address...

PC4 : 10.2.4.10 255.255.255.0 gateway 10.2.4.1

PC4> save

Saving startup configuration to startup.vpc

. done

PC4> show ip

NAME : PC4[1]

IP/MASK : 10.2.4.10/24

GATEWAY : 10.2.4.1

DNS :

MAC : 00:50:79:66:68:03

LPORT : 27310

RHOST:PORT : 127.0.0.1:27311

MTU : 1500

PC5> ip 10.1.5.10/24 10.1.5.1

Checking for duplicate address...

PC5 : 10.1.5.10 255.255.255.0 gateway 10.1.5.1

PC5> save

Saving startup configuration to startup.vpc

. done

PC5> show ip

NAME : PC5[1]

IP/MASK : 10.1.5.10/24

GATEWAY : 10.1.5.1
 DNS :
 MAC : 00:50:79:66:68:04
 LPORT : 27312
 RHOST:PORT : 127.0.0.1:27313
 MTU : 1500

R1#enable

R1#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Prot ocol
FastEthernet0/0	unassigned	YES	unset	administratively down	down
FastEthernet1/0	unassigned	YES	unset	administratively down	down

R1#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
 S - Switch, H - Host, I - IGMP, r - Repeater

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
-----------	---------------	---------	------------	----------	---------

R1#enable

R1#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#interface FastEthernet0/0

R1(config-if)#no shutdown

R1(config-if)#

*Mar 1 00:28:05.935: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t o up

*Mar 1 00:28:06.935: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern et0/0, changed state to up

R1(config-if)#exit

R1(config)#interface FastEthernet1/0

R1(config-if)#no shutdown

R1(config-if)#

*Mar 1 00:28:23.831: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up

*Mar 1 00:28:24.831: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R1(config-if)#exit

R1(config)#end

R1#

*Mar 1 00:28:31.191: %SYS-5-CONFIG_I: Configured from console by console

R1#

R1#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
vIOS-L2-01	Fas 1/0	124	R S I	IOSv	Gig 0/0
R5	Fas 0/0	150	R S I	3640	Fas 2/0

R1#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	up	up
FastEthernet1/0	unassigned	YES	unset	up	up

R6#enable

R6#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R6(config)#interface FastEthernet0/0

R6(config-if)#no shutdown

R6(config-if)#

*Mar 1 00:29:09.451: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:29:10.451: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R6(config-if)#exit

R6(config)#interface FastEthernet1/0

R6(config-if)#no shutdown

R6(config-if)#

*Mar 1 00:29:26.943: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up

*Mar 1 00:29:27.943: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R6(config-if)#exit

R6(config)#interface FastEthernet2/0

R6(config-if)#no shutdown

R6(config-if)#exit

*Mar 1 00:29:41.495: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up

*Mar 1 00:29:42.495: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up

R6(config-if)#exit

R6(config)#end

R6#

*Mar 1 00:29:49.919: %SYS-5-CONFIG_I: Configured from console by console

R6#

R6#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
R2	Fas 0/0	138	R S I	3640	Fas 1/0

R7 Fas 2/0 144 R S I 3640 Fas 2/0

R6#show ip interface brief

Interface

R2#show running-config | section interface

interface FastEthernet0/0

ip address 192.168.123.2 255.255.255.0

duplex auto

speed auto

interface FastEthernet1/0

ip address 192.168.26.2 255.255.255.0

duplex auto

speed auto

R2#

R5#show running-config | section interface

interface FastEthernet0/0

ip address 192.168.2.1 255.255.255.0

duplex auto

speed auto

interface FastEthernet1/0

ip address 192.168.45.5 255.255.255.0

duplex auto

speed auto

interface FastEthernet2/0

ip address 192.168.51.5 255.255.255.0

duplex auto

speed auto

PC1> ip 192.168.1.10/24 192.168.1.1

Checking for duplicate address...

PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC1> save

Saving startup configuration to startup.vpc

. done

PC1> show ip

NAME : PC1[1]

IP/MASK : 192.168.1.10/24

GATEWAY : 192.168.1.1

DNS :

MAC : 00:50:79:66:68:00

LPORT : 27304

RHOST:PORT : 127.0.0.1:27305

MTU : 1500

PC1> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=255 time=71.088 ms

84 bytes from 192.168.1.1 icmp_seq=2 ttl=255 time=11.271 ms

84 bytes from 192.168.1.1 icmp_seq=3 ttl=255 time=5.589 ms

84 bytes from 192.168.1.1 icmp_seq=4 ttl=255 time=7.372 ms

84 bytes from 192.168.1.1 icmp_seq=5 ttl=255 time=10.262 ms

PC2> ip 192.168.2.10/24 192.168.2.1

Checking for duplicate address...

PC2 : 192.168.2.10 255.255.255.0 gateway 192.168.2.1

PC2> save

Saving startup configuration to startup.vpc

. done

PC2> show ip

NAME : PC2[1]
IP/MASK : 192.168.2.10/24
GATEWAY : 192.168.2.1
DNS :
MAC : 00:50:79:66:68:01
LPORT : 27306
RHOST:PORT : 127.0.0.1:27307
MTU : 1500

PC2> ping 192.168.2.1

84 bytes from 192.168.2.1 icmp_seq=1 ttl=255 time=60.670 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=255 time=14.898 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=255 time=13.249 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=255 time=3.970 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=255 time=4.362 ms

PC3> ip 192.168.3.10/24 192.168.3.1

Checking for duplicate address...

PC3 : 192.168.3.10 255.255.255.0 gateway 192.168.3.1

PC3> save

Saving startup configuration to startup.vpc

. done

PC3> show ip

NAME : PC3[1]
IP/MASK : 192.168.3.10/24

GATEWAY : 192.168.3.1
DNS :
MAC : 00:50:79:66:68:02
LPORT : 27308
RHOST:PORT : 127.0.0.1:27309
MTU : 1500

PC4> ip 192.168.4.10/24 192.168.4.1

Checking for duplicate address...

PC4 : 192.168.4.10 255.255.255.0 gateway 192.168.4.1

PC4> save

Saving startup configuration to startup.vpc

. done

PC4> show ip

NAME : PC4[1]
IP/MASK : 192.168.4.10/24
GATEWAY : 192.168.4.1
DNS :
MAC : 00:50:79:66:68:03
LPORT : 27310
RHOST:PORT : 127.0.0.1:27311
MTU : 1500

PC4> ping 192.168.4.1

84 bytes from 192.168.4.1 icmp_seq=1 ttl=255 time=62.119 ms

84 bytes from 192.168.4.1 icmp_seq=2 ttl=255 time=15.014 ms

84 bytes from 192.168.4.1 icmp_seq=3 ttl=255 time=7.825 ms

84 bytes from 192.168.4.1 icmp_seq=4 ttl=255 time=4.057 ms
84 bytes from 192.168.4.1 icmp_seq=5 ttl=255 time=12.984 ms

PC5> ip 192.168.5.10/24 192.168.5.1

Checking for duplicate address...

PC5 : 192.168.5.10 255.255.255.0 gateway 192.168.5.1

PC5> save

Saving startup configuration to startup.vpc

. done

PC5> show ip

NAME : PC5[1]
IP/MASK : 192.168.5.10/24
GATEWAY : 192.168.5.1
DNS :
MAC : 00:50:79:66:68:04
LPORT : 27312
RHOST:PORT : 127.0.0.1:27313
MTU : 1500

PC5> ping 192.168.5.1

84 bytes from 192.168.5.1 icmp_seq=1 ttl=255 time=60.943 ms
84 bytes from 192.168.5.1 icmp_seq=2 ttl=255 time=12.250 ms
84 bytes from 192.168.5.1 icmp_seq=3 ttl=255 time=6.847 ms
84 bytes from 192.168.5.1 icmp_seq=4 ttl=255 time=16.459 ms
84 bytes from 192.168.5.1 icmp_seq=5 ttl=255 time=10.066 ms

R1#enable

R1#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#interface FastEthernet0/0

R1(config-if)#ip address 192.168.51.1 255.255.255.0

R1(config-if)# no shutdown

R1(config-if)#exit

R1(config)#interface FastEthernet1/0

R1(config-if)#ip address 192.168.123.1 255.255.255.0

R1(config-if)#no shutdown

R1(config-if)#exit

R1(config)#end

R1#

*Mar 1 00:45:30.499: %SYS-5-CONFIG_I: Configured from console by console

R1#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

R1#

R1#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.51.1	YES	manual	up	up
FastEthernet1/0	192.168.123.1	YES	manual	up	up

R6#enable

R6#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R6(config)#interface FastEthernet0/0

R6(config-if)#ip address 192.168.26.6 255.255.255.0

R6(config-if)#no shutdown

R6(config-if)#exit

R6(config)#interface FastEthernet1/0

```
R6(config-if)#ip address 192.168.3.1 255.255.255.0
```

```
R6(config-if)#no shutdown
```

```
R6(config-if)#exit
```

```
R6(config)#interface FastEthernet2/0
```

```
R6(config-if)#ip address 192.168.67.6 255.255.255.0
```

```
R6(config-if)#no shutdown
```

```
R6(config-if)#exit
```

```
R6(config)#end
```

```
R6#
```

```
*Mar  1 00:33:49.751: %SYS-5-CONFIG_I: Configured from console by console
```

```
R6#copy running-config startup-config
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

```
[OK]
```

```
R6#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.26.6	YES	manual	up	up
FastEthernet1/0	192.168.3.1	YES	manual	up	up
FastEthernet2/0	192.168.67.6	YES	manual	up	up

```
PC3> ping 192.168.3.1
```

```
84 bytes from 192.168.3.1 icmp_seq=1 ttl=255 time=70.863 ms
```

```
84 bytes from 192.168.3.1 icmp_seq=2 ttl=255 time=10.423 ms
```

```
84 bytes from 192.168.3.1 icmp_seq=3 ttl=255 time=11.471 ms
```

```
84 bytes from 192.168.3.1 icmp_seq=4 ttl=255 time=5.879 ms
```

```
84 bytes from 192.168.3.1 icmp_seq=5 ttl=255 time=7.643 ms
```

```
PC1> ping 192.168.1.1
```

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=255 time=9.225 ms
```

84 bytes from 192.168.1.1 icmp_seq=2 ttl=255 time=5.914 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=255 time=5.106 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=255 time=7.594 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=255 time=2.985 ms

PC2> ping 192.168.2.1

84 bytes from 192.168.2.1 icmp_seq=1 ttl=255 time=9.030 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=255 time=14.524 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=255 time=6.812 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=255 time=16.282 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=255 time=6.012 ms

PC3> ping 192.168.3.1

84 bytes from 192.168.3.1 icmp_seq=1 ttl=255 time=10.269 ms
84 bytes from 192.168.3.1 icmp_seq=2 ttl=255 time=14.214 ms
84 bytes from 192.168.3.1 icmp_seq=3 ttl=255 time=2.964 ms
84 bytes from 192.168.3.1 icmp_seq=4 ttl=255 time=13.007 ms
84 bytes from 192.168.3.1 icmp_seq=5 ttl=255 time=7.927 ms

PC4> ping 192.168.4.1

84 bytes from 192.168.4.1 icmp_seq=1 ttl=255 time=10.173 ms
84 bytes from 192.168.4.1 icmp_seq=2 ttl=255 time=6.852 ms
84 bytes from 192.168.4.1 icmp_seq=3 ttl=255 time=4.480 ms
84 bytes from 192.168.4.1 icmp_seq=4 ttl=255 time=14.636 ms
84 bytes from 192.168.4.1 icmp_seq=5 ttl=255 time=9.234 ms

PC5> ping 192.168.5.1

84 bytes from 192.168.5.1 icmp_seq=1 ttl=255 time=12.243 ms
84 bytes from 192.168.5.1 icmp_seq=2 ttl=255 time=15.173 ms

84 bytes from 192.168.5.1 icmp_seq=3 ttl=255 time=6.524 ms

84 bytes from 192.168.5.1 icmp_seq=4 ttl=255 time=6.663 ms

84 bytes from 192.168.5.1 icmp_seq=5 ttl=255 time=13.299 ms

Устр ойство	Инте рфейс	IP	Сеть
PC1	eth0	192.168.1.1 0/24	192.168.1.0 /24
R4	Fa0/ 0	192.168.1.1 /24	192.168.1.0 /24
R4	Fa1/ 0	192.168.45. 4/24	192.168.45. 0/24
R5	Fa1/ 0	192.168.45. 5/24	192.168.45. 0/24
PC2	eth0	192.168.2.1 0/24	192.168.2.0 /24
R5	Fa0/ 0	192.168.2.1 /24	192.168.2.0 /24
R5	Fa2/ 0	192.168.51. 5/24	192.168.51. 0/24
R1	Fa0/ 0	192.168.51. 1/24	192.168.51. 0/24
R1	Fa1/ 0	192.168.123 .1/24	192.168.123 .0/24
R2	Fa0/ 0	192.168.123 .2/24	192.168.123 .0/24
R3	Fa0/ 0	192.168.123 .3/24	192.168.123 .0/24

Устр ойство	Инте рфейс	IP	Сеть
R2	Fa1/ 0	192.168.26. 2/24	192.168.26. 0/24
R6	Fa0/ 0	192.168.26. 6/24	192.168.26. 0/24
R6	Fa1/ 0	192.168.3.1 /24	192.168.3.0 /24
PC3	eth0	192.168.3.1 0/24	192.168.3.0 /24
R6	Fa2/ 0	192.168.67. 6/24	192.168.67. 0/24
R7	Fa2/ 0	192.168.67. 7/24	192.168.67. 0/24
R7	Fa0/ 0	192.168.37. 7/24	192.168.37. 0/24
R3	Fa1/ 0	192.168.37. 3/24	192.168.37. 0/24
R7	Fa1/ 0	192.168.4.1 /24	192.168.4.0 /24
PC4	eth0	192.168.4.1 0/24	192.168.4.0 /24
R3	Fa2/ 0	192.168.38. 3/24	192.168.38. 0/24
R8	Fa0/ 0	192.168.38. 8/24	192.168.38. 0/24
R8	Fa1/ 0	192.168.5.1 /24	192.168.5.0 /24

Устр ойство	Инте рфейс	IP	Сеть
	0	/24	/24
PC5	eth0	192.168.5.1 0/24	192.168.5.0 /24

ЗАДАНИЕ 2

2) Настроить протокол динамической маршрутизации RIP v2 для области, указанной на схеме schema-lab5.

```
R4#enable
```

```
R4#configure terminal
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
R4(config)#router rip
```

```
R4(config-router)#version 2
```

```
R4(config-router)#no auto-summary
```

```
R4(config-router)#network 192.168.1.0
```

```
R4(config-router)#network 192.168.45.0
```

```
R4(config-router)#passive-interface FastEthernet0/0
```

```
R4(config-router)#exit
```

```
R4(config)#end
```

```
R4#
```

```
*Mar  1 00:55:05.083: %SYS-5-CONFIG_I: Configured from console by console
```

```
R4#copy running-config startup-config
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

```
[OK]
```

```
R4#
```

```
R5#enable
```

```
R5#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.2.1	YES	NVRAM	up	up
FastEthernet1/0	192.168.45.5	YES	NVRAM	up	up
FastEthernet2/0	192.168.51.5	YES	NVRAM	up	up

```
R5#show cdp neighbors
```

```
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
```

S - Switch, H - Host, I - IGMP, r - Repeater

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
R4	Fas 1/0	159	R S I	3640	Fas 1/0

R5#show running-config | section interface

interface FastEthernet0/0

ip address 192.168.2.1 255.255.255.0

duplex auto

speed auto

interface FastEthernet1/0

ip address 192.168.45.5 255.255.255.0

duplex auto

speed auto

interface FastEthernet2/0

ip address 192.168.51.5 255.255.255.0

duplex auto

speed auto

R5#enable

R5#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R5(config)#router rip

R5(config-router)# version 2

R5(config-router)#no auto-summary

R5(config-router)#network 192.168.2.0

R5(config-router)#network 192.168.45.0

R5(config-router)#network 192.168.51.0

R5(config-router)#passive-interface FastEthernet0/0

R5(config-router)#exit

R5(config)#end

R5#

*Mar 1 00:56:31.115: %SYS-5-CONFIG_I: Configured from console by console

R5#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

R5#

R1#enable

R1#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#router rip

R1(config-router)#version 2

R1(config-router)#no auto-summary

R1(config-router)#network 192.168.51.0

R1(config-router)#exit

R1(config)#end

R1#

*Mar 1 01:01:17.383: %SYS-5-CONFIG_I: Configured from console by console

R1#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

R1#

R1#show ip protocols

Routing Protocol is "rip"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Sending updates every 30 seconds, next due in 4 seconds

Invalid after 180 seconds, hold down 180, flushed after 240

Redistributing: rip

Default version control: send version 2, receive version 2

Interface	Send	Recv	Triggered RIP	Key-chain
FastEthernet0/0	2	2		

Automatic network summarization is not in effect

Maximum path: 4

Routing for Networks:

192.168.51.0

Routing Information Sources:

Gateway	Distance	Last Update
192.168.51.5	120	00:00:24

Distance: (default is 120)

R1#show ip route rip

R 192.168.45.0/24 [120/1] via 192.168.51.5, 00:00:31, FastEthernet0/0

R 192.168.1.0/24 [120/2] via 192.168.51.5, 00:00:31, FastEthernet0/0

R 192.168.2.0/24 [120/1] via 192.168.51.5, 00:00:31, FastEthernet0/0

R4#show ip protocols

Routing Protocol is "rip"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Sending updates every 30 seconds, next due in 19 seconds

Invalid after 180 seconds, hold down 180, flushed after 240

Redistributing: rip

Default version control: send version 2, receive version 2

Interface	Send	Recv	Triggered RIP	Key-chain
FastEthernet1/0	2	2		

Automatic network summarization is not in effect

Maximum path: 4

Routing for Networks:

192.168.1.0

192.168.45.0

Passive Interface(s):

FastEthernet0/0

Routing Information Sources:

Gateway	Distance	Last Update
192.168.45.5	120	00:00:19

Distance: (default is 120)

R4#show ip route rip

R 192.168.51.0/24 [120/1] via 192.168.45.5, 00:00:26, FastEthernet1/0

R 192.168.2.0/24 [120/

R5#show ip protocols

Routing Protocol is "rip"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Sending updates every 30 seconds, next due in 22 seconds

Invalid after 180 seconds, hold down 180, flushed after 240

Redistributing: rip

Default version control: send version 2, receive version 2

Interface	Send	Recv	Triggered RIP	Key-chain
FastEthernet1/0	2	2		
FastEthernet2/0	2	2		

Automatic network summarization is not in effect

Maximum path: 4

Routing for Networks:

192.168.2.0

192.168.45.0

192.168.51.0

Passive Interface(s):

FastEthernet0/0

Routing Information Sources:

Gateway	Distance	Last Update
192.168.45.4	120	00:00:38

Distance: (default is

R4#terminal length 0

R4#show ip route rip

R 192.168.51.0/24 [120/1] via 192.168.45.5, 00:00:40, FastEthernet1/0

R 192.168.2.0/24 [120/1] via 192.168.45.5, 00:00:40, FastEthernet1/0

R5#terminal length 0

R5#show ip route rip

R 192.168.1.0/24 [120/1] via 192.168.45.4, 00:00:24, FastEthernet1/0

PC1> ping 192.168.2.10

84 bytes from 192.168.2.10 icmp_seq=1 ttl=62 time=59.224 ms

84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=33.620 ms

84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=40.513 ms

84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=32.289 ms

84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=24.870 ms

PC2> ping 192.168.1.10

84 bytes from 192.168.1.10 icmp_seq=1 ttl=62 time=25.607 ms

84 bytes from 192.168.1.10 icmp_seq=2 ttl=62 time=34.745 ms

84 bytes from 192.168.1.10 icmp_seq=3 ttl=62 time=30.647 ms

84 bytes from 192.168.1.10 icmp_seq=4 ttl=62 time=29.283 ms

84 bytes from 192.168.1.10 icmp_seq=5 ttl=62 time=38.766 ms

ЗАДАНИЕ 3

- 3) Настроить протокол динамической маршрутизации OSPF для зон 0, 1,
2. Зону 1 настроить как полностью (nb!) тупиковую.

```
R1#enable
```

```
R1#configure terminal
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
R1(config)#router ospf 1
```

```
R1(config-router)#router-id 1.1.1.1
```

```
R1(config-router)#network 192.168.123.0 0.0.0.255 area 0
```

```
R1(config-router)#exit
```

```
R1(config)#end
```

```
R1#
```

```
*Mar  1 01:19:27.955: %SYS-5-CONFIG_I: Configured from console by console
```

```
R1#copy running-config startup-config
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

```
[OK]
```

```
R1#enable
```

```
R1#configure terminal
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
R1(config)#router ospf 1
```

```
R1(config-router)#router-id 1.1.1.1
```

```
R1(config-router)#network 192.168.123.0 0.0.0.255 area 0
```

```
R1(config-router)#exit
```

```
R1(config)#end
```

```
R1#
```

```
*Mar  1 01:19:27.955: %SYS-5-CONFIG_I: Configured from console by console
```

```
R1#copy running-config startup-config
```

Destination filename [startup-config]?

Building configuration...

[OK]

R1#

*Mar 1 01:20:38.459: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on
FastEthernet1/0 from LOADING to FULL, Loading Done

R1#

R2#enable

R2#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#router ospf 1

R2(config-router)#router-id 2.2.2.2

R2(config-router)#network 192.168.123.0 0.0.0.255 area 0

R2(config-router)#exit

R2(config)#end

R2#

*Mar 1 01:18:18.375: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on
FastEthernet0/0 from LOADING to FULL, Loading Done

*Mar 1 01:18:19.251: %SYS-5-CONFIG_I: Configured from console by
console

R2#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

R2#

R3#enable

R3#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R3(config)#router ospf 1

R3(config-router)#router-id 3.3.3.3

```

R3(config-router)#network 192.168.123.0 0.0.0.255 area 0
R3(config-router)#exit
R3(config)#
*Mar  1 01:21:38.547: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on
FastEthernet0/0 from LOADING to FULL, Loading Done
*Mar  1 01:21:38.623: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on
FastEthernet0/0 from LOADING to FULL, Loading Done
R3(config)#end
R3#
*Mar  1 01:21:44.979: %SYS-5-CONFIG_I: Configured from console by
console
R3#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R3#

```

```

R1#terminal length 0
R1#show ip ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	FULL/BDR	00:00:38	192.168.123.2	FastEthernet1/0
3.3.3.3		1 FULL/DROTHER	00:00:33	192.168.123.3	FastEthernet1/0

```

R2#terminal length 0
R2#show ip ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	1	FULL/DR	00:00:32	192.168.123.1	FastEthernet0/0
3.3.3.3		1 FULL/DROTHER	00:00:37	192.168.123.3	FastEthernet0/0

R2#

R3#terminal length 0

R3#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	1	FULL/DR	00:00:34	192.168.123.1	FastEthernet0/0
2.2.2.2	1	FULL/BDR	00:00:34	192.168.123.2	FastEthernet0/0

R3#enable

R3#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R3(config)#router ospf 1

R3(config-router)#network 192.168.38.0 0.0.0.255 area 1

R3(config-router)#area 1 stub no-summary

R3(config-router)#exit

R3(config)#end

R3#

*Mar 1 01:25:45.731: %SYS-5-CONFIG_I: Configured from console by console

R3#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

R3#

R8#enable

R8#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R8(config)#router ospf 1

R8(config-router)#router-id 8.8.8.8

R8(config-router)#network 192.168.38.0 0.0.0.255 area 1

R8(config-router)#network 192.168.5.0 0.0.0.255 area 1

```

R8(config-router)#area 1 stub
R8(config-router)#passive-interface FastEthernet1/0
R8(config-router)#
*Mar  1 01:21:17.499: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on
FastEthernet0/0 from LOADING to FULL, Loading Done
R8(config-router)#exit
R8(config)#end
R8#
*Mar  1 01:21:29.475: %SYS-5-CONFIG_I: Configured from console by
console
R8#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]

```

```

R8#terminal length 0
*Mar  1 01:21:58.007: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on
FastEthernet0/0 from FULL to DOWN, Neighbor Down: Dead timer expired
R8#terminal length 0
R8#show ip ospf neighbor

R8#show ip route ospf

```

```

R2#enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)# network 192.168.26.0 0.0.0.255 area 2
R2(config-router)#exit
R2(config)#end
R2#

```

*Mar 1 01:28:53.355: %SYS-5-CONFIG_I: Configured from console by console

R2#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

R3#enable

R3#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R3(config)#router ospf 1

R3(config-router)#network 192.168.37.0 0.0.0.255 area 2

R3(config-router)#exit

R3(config)#end

R3#

*Mar 1 01:35:07.039: %SYS-5-CONFIG_I: Configured from console by console

R3#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

R3#

R6#enable

R6#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R6(config)#router ospf 1

R6(config-router)#router-id 6.6.6.6

R6(config-router)#network 192.168.26.0 0.0.0.255 area 2

R6(config-router)#network 192.168.3.0 0.0.0.255 area 2

R6(config-router)#

*Mar 1 01:29:50.783: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on FastEthernet0/0 from LOADING to FULL, Loading Done

R6(config-router)#network 192.168.67.0 0.0.0.255 area 2

R6(config-router)#passive-interface FastEthernet1/0

R6(config-router)#exit

R6(config)#end

R6#

*Mar 1 01:30:19.491: %SYS-5-CONFIG_I: Configured from console by console

R6#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

R6#

R7#enable

R7#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R7(config)#router ospf 1

R7(config-router)#router-id 7.7.7.7

R7(config-router)#network 192.168.37.0 0.0.0.255 area 2

R7(config-router)#

*Mar 1 01:41:43.475: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on FastEthernet0/0 from LOADING to FULL, Loading Done

R7(config-router)#network 192.168.4.0 0.0.0.255 area 2

R7(config-router)#network 192.168.67.0 0.0.0.255 area 2

R7(config-router)#passive-interface FastEthernet1/0

R7(config-router)#

*Mar 1 01:42:07.255: %OSPF-5-ADJCHG: Process 1, Nbr 6.6.6.6 on FastEthernet2/0 from LOADING to FULL, Loading Done

R7(config-router)#exit

R7(config)#end

R7#

*Mar 1 01:42:16.103: %SYS-5-CONFIG_I: Configured from console by console

R7#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

R7#

R2#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1		1 FULL/DROTHER	00:00:35	192.168.123.1	FastEthernet0/0

3.3.3.3	1	FULL/DR	00:00:31	192.168.123.3	FastEthernet0/0
---------	---	---------	----------	---------------	-----------------

6.6.6.6	1	FULL/BDR	00:00:39	192.168.26.6	FastEthernet1/0
---------	---	----------	----------	--------------	-----------------

R2#

R3#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1		1 FULL/DROTHER	00:00:33	192.168.123.1	FastEthernet0/0

2.2.2.2	1	FULL/BDR	00:00:34	192.168.123.2	FastEthernet0/0
---------	---	----------	----------	---------------	-----------------

8.8.8.8	1	FULL/DR	00:00:32	192.168.38.8	FastEthernet2/0
---------	---	---------	----------	--------------	-----------------

7.7.7.7	1	FULL/BDR	00:00:39	192.168.37.7	FastEthernet1/0
---------	---	----------	----------	--------------	-----------------

R3#

R6#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
7.7.7.7	1	FULL/BDR	00:00:32	192.168.67.7	FastEthernet2/0

2.2.2.2 1 FULL/DR 00:00:35 192.168.26.2 FastEthernet0/0
R6#

R7#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
6.6.6.6	1	FULL/DR	00:00:35	192.168.67.6	FastEthernet2/0
3.3.3.3	1	FULL/DR	00:00:31	192.168.37.3	FastEthernet0/0

R8#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	1	FULL/BDR	00:00:30	192.168.38.3	FastEthernet0/0

R8#terminal length 0

R8#show ip route ospf

O*IA 0.0.0.0/0 [110/2] via 192.168.38.3, 00:16:06, FastEthernet0/0

R8#show ip route ospf

O*IA 0.0.0.0/0 [110/2] via 192.168.38.3, 00:16:06, FastEthernet0/0

R8#show ip ospf

Routing Process "ospf 1" with ID 8.8.8.8

Start time: 01:20:42.720, Time elapsed: 00:27:46.172

Supports only single TOS(TOS0) routes

Supports opaque LSA

Supports Link-local Signaling (LLS)

Supports area transit capability

Router is not originating router-LSAs with maximum metric

Initial SPF schedule delay 5000 msecs

Minimum hold time between two consecutive SPF's 10000 msecs

Maximum wait time between two consecutive SPF's 10000 msecs

Incremental-SPF disabled

Minimum LSA interval 5 secs

Minimum LSA arrival 1000 msec
LSA group pacing timer 240 sec
Interface flood pacing timer 33 msec
Retransmission pacing timer 66 msec
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 0 normal 1 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0

Area 1

Number of interfaces in this area is 2
It is a stub area
Area has no authentication
SPF algorithm last executed 00:16:39.296 ago
SPF algorithm executed 7 times
Area ranges are
Number of LSA 4. Checksum Sum 0x029623
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

R8#

ЗАДАНИЕ 4

4) Настроить редистрибуцию маршрутов между протоколами RIP v2 и OSPF.

```
R1#enable
```

```
R1#configure terminal
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
R1(config)#router ospf 1
```

```
R1(config-router)#redistribute rip subnets
```

```
R1(config-router)#exit
```

```
R1(config)#end
```

```
R1#
```

```
*Mar  1 01:58:19.219: %SYS-5-CONFIG_I: Configured from console by console
```

```
R1#copy running-config startup-config
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

```
[OK]
```

```
R1#
```

```
R1#configure terminal
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
R1(config)#router rip
```

```
R1(config-router)#redistribute ospf 1 metric 2
```

```
R1(config-router)#exit
```

```
R1(config)#end
```

```
R1#
```

```
*Mar  1 02:00:06.903: %SYS-5-CONFIG_I: Configured from console by console
```

```
R1#copy running-config startup-config
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

[OK]

R1#

R2#terminal length 0

R2#show ip route ospf

O E2 192.168.45.0/24 [110/20] via 192.168.123.1, 00:03:28, FastEthernet0/0
O IA 192.168.38.0/24 [110/2] via 192.168.123.3, 00:03:38, FastEthernet0/0
O 192.168.4.0/24 [110/3] via 192.168.26.6, 00:12:14, FastEthernet1/0
O IA 192.168.5.0/24 [110/3] via 192.168.123.3, 00:03:38, FastEthernet0/0
O 192.168.67.0/24 [110/2] via 192.168.26.6, 00:12:14, FastEthernet1/0
O 192.168.37.0/24 [110/3] via 192.168.26.6, 00:12:14, FastEthernet1/0
O E2 192.168.51.0/24 [110/20] via 192.168.123.1, 00:03:28, FastEthernet0/0
O E2 192.168.1.0/24 [110/20] via 192.168.123.1, 00:03:28, FastEthernet0/0
O E2 192.168.2.0/24 [110/20] via 192.168.123.1, 00:03:28, FastEthernet0/0
O 192.168.3.0/24 [110/2] via 192.168.26.6, 00:12:14, FastEthernet1/0

R3#terminal length 0

R3#show ip route ospf

O E2 192.168.45.0/24 [110/20] via 192.168.123.1, 00:04:34, FastEthernet0/0
O 192.168.26.0/24 [110/3] via 192.168.37.7, 00:13:15, FastEthernet1/0
O 192.168.4.0/24 [110/2] via 192.168.37.7, 00:13:15, FastEthernet1/0
O 192.168.5.0/24 [110/2] via 192.168.38.8, 00:16:55, FastEthernet2/0
O 192.168.67.0/24 [110/2] via 192.168.37.7, 00:13:15, FastEthernet1/0
O E2 192.168.51.0/24 [110/20] via 192.168.123.1, 00:04:34, FastEthernet0/0
O E2 192.168.1.0/24 [110/20] via 192.168.123.1, 00:04:34, FastEthernet0/0
O E2 192.168.2.0/24 [110/20] via 192.168.123.1, 00:04:34, FastEthernet0/0
O 192.168.3.0/24 [110/3] via 192.168.37.7, 00:13:15, FastEthernet1/0

R3#

R5#terminal length 0

R5#show ip route rip

R 192.168.123.0/24 [120/2] via 192.168.51.1, 00:00:15, FastEthernet2/0
R 192.168.26.0/24 [120/2] via 192.168.51.1, 00:00:15, FastEthernet2/0

```
R 192.168.38.0/24 [120/2] via 192.168.51.1, 00:00:15, FastEthernet2/0
R 192.168.4.0/24 [120/2] via 192.168.51.1, 00:00:15, FastEthernet2/0
R 192.168.5.0/24 [120/2] via 192.168.51.1, 00:00:15, FastEthernet2/0
R 192.168.67.0/24 [120/2] via 192.168.51.1, 00:00:15, FastEthernet2/0
R 192.168.37.0/24 [120/2] via 192.168.51.1, 00:00:15, FastEthernet2/0
R 192.168.1.0/24 [120/1] via 192.168.45.4, 00:00:04, FastEthernet1/0
R 192.168.3.0/24 [120/2] via 192.168.51.1, 00:00:15, FastEthernet2/0
R5#
```

```
R4#terminal length 0
```

```
R4#show ip route rip
```

```
R 192.168.123.0/24 [120/3] via 192.168.45.5, 00:00:09, FastEthernet1/0
R 192.168.26.0/24 [120/3] via 192.168.45.5, 00:00:09, FastEthernet1/0
R 192.168.38.0/24 [120/3] via 192.168.45.5, 00:00:09, FastEthernet1/0
R 192.168.4.0/24 [120/3] via 192.168.45.5, 00:00:09, FastEthernet1/0
R 192.168.5.0/24 [120/3] via 192.168.45.5, 00:00:09, FastEthernet1/0
R 192.168.67.0/24 [120/3] via 192.168.45.5, 00:00:09, FastEthernet1/0
R 192.168.37.0/24 [120/3] via 192.168.45.5, 00:00:09, FastEthernet1/0
R 192.168.51.0/24 [120/1] via 192.168.45.5, 00:00:09, FastEthernet1/0
R 192.168.2.0/24 [120/1] via 192.168.45.5, 00:00:09, FastEthernet1/0
R 192.168.3.0/24 [120/3] via 192.168.45.5, 00:00:09, FastEthernet1/0
```

```
R8#show ip route ospf
```

```
O*IA 0.0.0.0/0 [110/2] via 192.168.38.3, 00:27:28, FastEthernet0/0
```

```
R8#
```

ЗАДАНИЕ 5

5) Проверить работоспособность маршрутизации, выполнив ping VPC "все между всеми" (nb!: в обе стороны).

```
PC1> ping 192.168.2.10
```

```
84 bytes from 192.168.2.10 icmp_seq=1 ttl=62 time=51.197 ms
```

```
84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=25.463 ms
```

```
84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=29.854 ms
```

```
84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=27.348 ms
```

```
84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=27.376 ms
```

```
PC1> ping 192.168.3.10
```

```
84 bytes from 192.168.3.10 icmp_seq=1 ttl=59 time=97.289 ms
```

```
84 bytes from 192.168.3.10 icmp_seq=2 ttl=59 time=92.141 ms
```

```
84 bytes from 192.168.3.10 icmp_seq=3 ttl=59 time=86.864 ms
```

```
84 bytes from 192.168.3.10 icmp_seq=4 ttl=59 time=95.664 ms
```

```
84 bytes from 192.168.3.10 icmp_seq=5 ttl=59 time=76.321 ms
```

```
PC1> ping 192.168.4.10
```

```
84 bytes from 192.168.4.10 icmp_seq=1 ttl=59 time=105.460 ms
```

```
84 bytes from 192.168.4.10 icmp_seq=2 ttl=59 time=104.847 ms
```

```
84 bytes from 192.168.4.10 icmp_seq=3 ttl=59 time=95.778 ms
```

```
84 bytes from 192.168.4.10 icmp_seq=4 ttl=59 time=68.423 ms
```

```
84 bytes from 192.168.4.10 icmp_seq=5 ttl=59 time=95.542 ms
```

```
PC1>
```

```
PC1> ping 192.168.5.10
```

84 bytes from 192.168.5.10 icmp_seq=1 ttl=59 time=123.634 ms
84 bytes from 192.168.5.10 icmp_seq=2 ttl=59 time=67.356 ms
84 bytes from 192.168.5.10 icmp_seq=3 ttl=59 time=107.940 ms
84 bytes from 192.168.5.10 icmp_seq=4 ttl=59 time=76.367 ms
84 bytes from 192.168.5.10 icmp_seq=5 ttl=59 time=91.264 ms

PC1>

PC2> ping 192.168.1.10

84 bytes from 192.168.1.10 icmp_seq=1 ttl=62 time=24.342 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=62 time=36.383 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=62 time=32.294 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=62 time=34.379 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=62 time=38.539 ms

PC2> ping 192.168.3.10

84 bytes from 192.168.3.10 icmp_seq=1 ttl=60 time=86.051 ms
84 bytes from 192.168.3.10 icmp_seq=2 ttl=60 time=79.129 ms
84 bytes from 192.168.3.10 icmp_seq=3 ttl=60 time=73.904 ms
84 bytes from 192.168.3.10 icmp_seq=4 ttl=60 time=66.366 ms
84 bytes from 192.168.3.10 icmp_seq=5 ttl=60 time=74.213 ms

PC2> ping 192.168.4.10

84 bytes from 192.168.4.10 icmp_seq=1 ttl=60 time=87.780 ms
84 bytes from 192.168.4.10 icmp_seq=2 ttl=60 time=70.304 ms
84 bytes from 192.168.4.10 icmp_seq=3 ttl=60 time=88.523 ms
84 bytes from 192.168.4.10 icmp_seq=4 ttl=60 time=85.235 ms

84 bytes from 192.168.4.10 icmp_seq=5 ttl=60 time=66.257 ms

PC2> ping 192.168.5.10

84 bytes from 192.168.5.10 icmp_seq=1 ttl=60 time=85.632 ms

84 bytes from 192.168.5.10 icmp_seq=2 ttl=60 time=76.251 ms

84 bytes from 192.168.5.10 icmp_seq=3 ttl=60 time=85.604 ms

84 bytes from 192.168.5.10 icmp_seq=4 ttl=60 time=82.592 ms

84 bytes from 192.168.5.10 icmp_seq=5 ttl=60 time=112.416 ms

PC3> ping 192.168.1.10

84 bytes from 192.168.1.10 icmp_seq=1 ttl=59 time=83.767 ms

84 bytes from 192.168.1.10 icmp_seq=2 ttl=59 time=101.962 ms

84 bytes from 192.168.1.10 icmp_seq=3 ttl=59 time=79.741 ms

84 bytes from 192.168.1.10 icmp_seq=4 ttl=59 time=92.840 ms

84 bytes from 192.168.1.10 icmp_seq=5 ttl=59 time=100.118 ms

PC3> ping 192.168.2.10

84 bytes from 192.168.2.10 icmp_seq=1 ttl=60 time=85.147 ms

84 bytes from 192.168.2.10 icmp_seq=2 ttl=60 time=149.050 ms

84 bytes from 192.168.2.10 icmp_seq=3 ttl=60 time=69.033 ms

84 bytes from 192.168.2.10 icmp_seq=4 ttl=60 time=103.502 ms

84 bytes from 192.168.2.10 icmp_seq=5 ttl=60 time=87.275 ms

ping 192.168.4.10

PC3> ping 192.168.4.10

84 bytes from 192.168.4.10 icmp_seq=1 ttl=62 time=44.674 ms

84 bytes from 192.168.4.10 icmp_seq=2 ttl=62 time=24.429 ms

84 bytes from 192.168.4.10 icmp_seq=3 ttl=62 time=35.354 ms

84 bytes from 192.168.4.10 icmp_seq=4 ttl=62 time=25.989 ms
84 bytes from 192.168.4.10 icmp_seq=5 ttl=62 time=25.978 ms

PC3> ping 192.168.5.10

84 bytes from 192.168.5.10 icmp_seq=1 ttl=60 time=88.195 ms
84 bytes from 192.168.5.10 icmp_seq=2 ttl=60 time=85.117 ms
84 bytes from 192.168.5.10 icmp_seq=3 ttl=60 time=68.587 ms
84 bytes from 192.168.5.10 icmp_seq=4 ttl=60 time=87.381 ms
84 bytes from 192.168.5.10 icmp_seq=5 ttl=60 time=73.660 ms

PC4> ping 192.168.1.10

84 bytes from 192.168.1.10 icmp_seq=1 ttl=59 time=90.236 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=59 time=88.324 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=59 time=86.811 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=59 time=87.781 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=59 time=94.266 ms

PC4> ping 192.168.2.10

84 bytes from 192.168.2.10 icmp_seq=1 ttl=60 time=80.133 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=60 time=56.329 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=60 time=73.404 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=60 time=145.726 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=60 time=95.923 ms

PC4> ping 192.168.3.10

84 bytes from 192.168.3.10 icmp_seq=1 ttl=62 time=33.306 ms
84 bytes from 192.168.3.10 icmp_seq=2 ttl=62 time=24.266 ms

84 bytes from 192.168.3.10 icmp_seq=3 ttl=62 time=24.952 ms
84 bytes from 192.168.3.10 icmp_seq=4 ttl=62 time=33.169 ms
84 bytes from 192.168.3.10 icmp_seq=5 ttl=62 time=25.681 ms

PC4> ping 192.168.5.10

84 bytes from 192.168.5.10 icmp_seq=1 ttl=61 time=49.498 ms
84 bytes from 192.168.5.10 icmp_seq=2 ttl=61 time=44.658 ms
84 bytes from 192.168.5.10 icmp_seq=3 ttl=61 time=42.947 ms
84 bytes from 192.168.5.10 icmp_seq=4 ttl=61 time=46.828 ms
84 bytes from 192.168.5.10 icmp_seq=5 ttl=61 time=37.325 ms

PC5> ping 192.168.1.10

84 bytes from 192.168.1.10 icmp_seq=1 ttl=59 time=101.401 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=59 time=84.709 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=59 time=95.950 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=59 time=79.003 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=59 time=85.858 ms

PC5> ping 192.168.2.10

84 bytes from 192.168.2.10 icmp_seq=1 ttl=60 time=85.595 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=60 time=87.273 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=60 time=66.345 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=60 time=56.117 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=60 time=67.339 ms

PC5> ping 192.168.3.10

84 bytes from 192.168.3.10 icmp_seq=1 ttl=60 time=90.818 ms
84 bytes from 192.168.3.10 icmp_seq=2 ttl=60 time=73.309 ms
84 bytes from 192.168.3.10 icmp_seq=3 ttl=60 time=56.817 ms
84 bytes from 192.168.3.10 icmp_seq=4 ttl=60 time=82.557 ms
84 bytes from 192.168.3.10 icmp_seq=5 ttl=60 time=68.835 ms

PC5> ping 192.168.4.10

84 bytes from 192.168.4.10 icmp_seq=1 ttl=61 time=59.260 ms
84 bytes from 192.168.4.10 icmp_seq=2 ttl=61 time=55.039 ms
84 bytes from 192.168.4.10 icmp_seq=3 ttl=61 time=43.611 ms
84 bytes from 192.168.4.10 icmp_seq=4 ttl=61 time=45.645 ms
84 bytes from 192.168.4.10 icmp_seq=5 ttl=61 time=42.972 ms

ЗАДАНИЕ 6

6) Перехватить в Wireshark сообщения протоколов RIP v2 и OSPF, идентифицировать их тип и содержание.

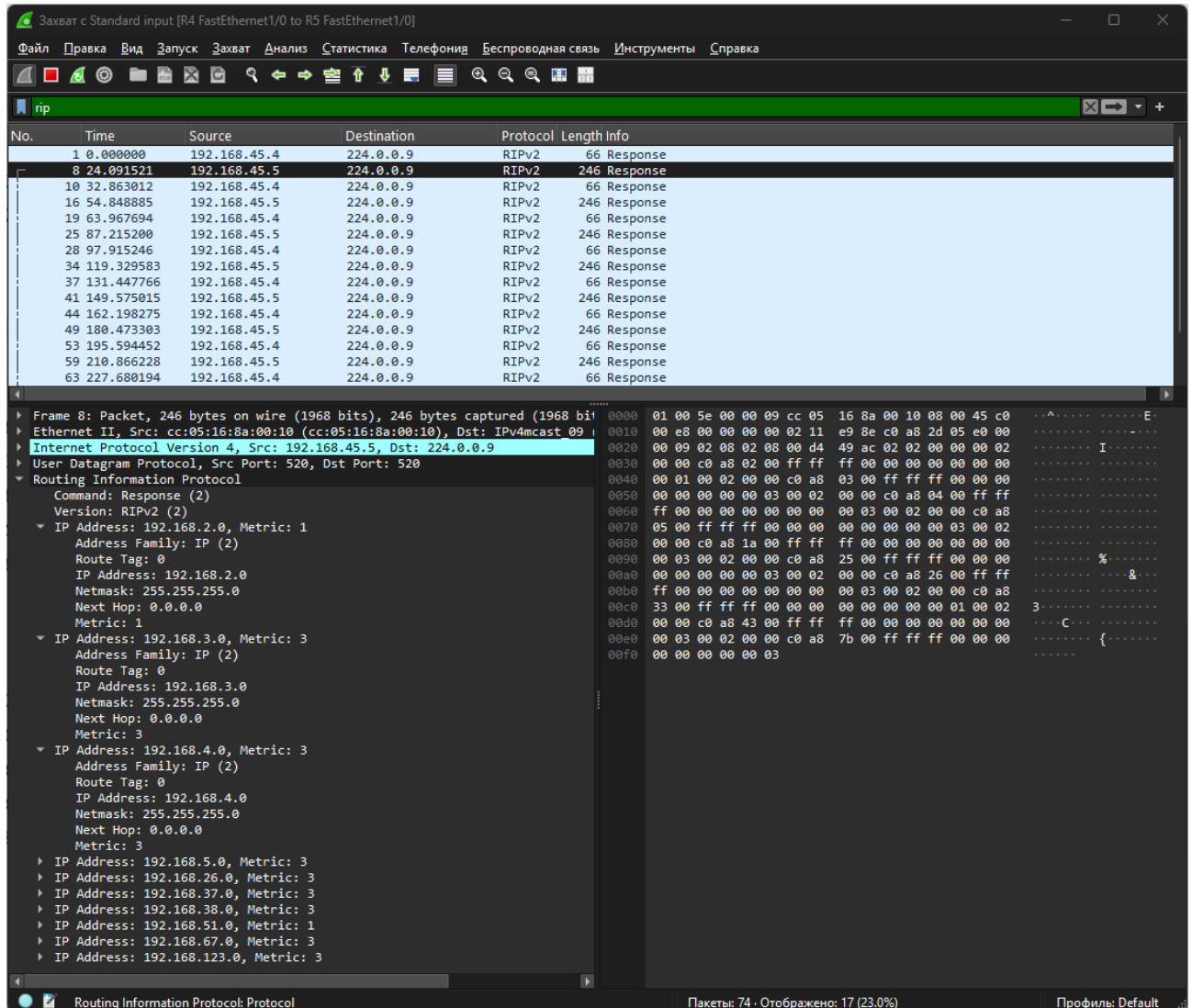


Рисунок 2 – Перехват сообщения протокола RIP v2 в Wireshark

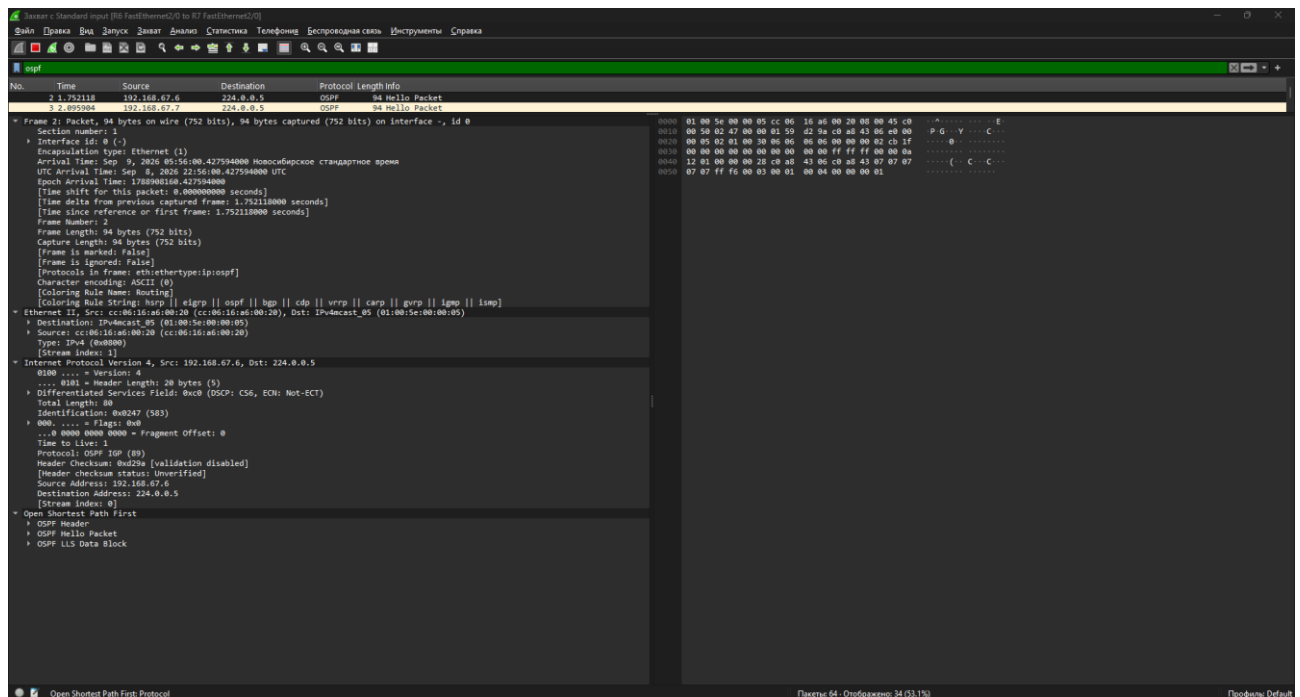


Рисунок 3 – Перехват сообщения OSPF Hello в Wireshark

R6#enable

R6#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R6(config)#interface FastEthernet2/0

R6(config-if)#shutdown

R6(config-if)#

*Mar 1 02:15:32.987: %OSPF-5-ADJCHG: Process 1, Nbr 7.7.7.7 on FastEthernet2/0 from FULL to DOWN, Neighbor Down: Interface down or detached

R6(config-if)#

*Mar 1 02:15:34.979: %LINK-5-CHANGED: Interface FastEthernet2/0, changed state to administratively down

*Mar 1 02:15:35.979: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to down

R6(config-if)#no shutdown

R6(config-if)#

*Mar 1 02:15:52.559: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up

*Mar 1 02:15:53.559: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up

R6(config-if)#end

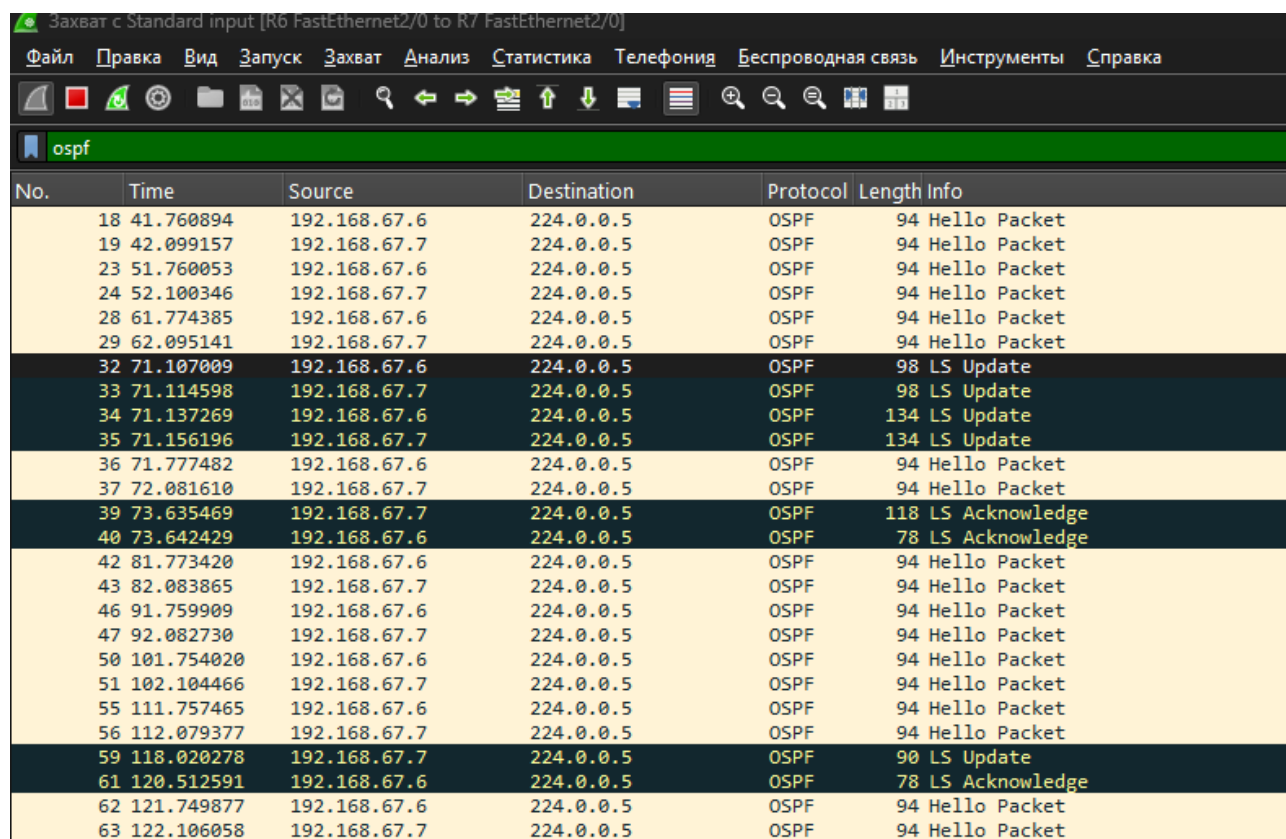
*Mar 1 02:15:55.719: %OSPF-5-ADJCHG: Process 1, Nbr 7.7.7.7 on FastEthernet2/0 from LOADING to FULL, Loading Done

R6(config-if)#end

R6#

*Mar 1 02:15:57.895: %SYS-5-CONFIG_I: Configured from console by console

R6#



No.	Time	Source	Destination	Protocol	Length	Info
18	41.760894	192.168.67.6	224.0.0.5	OSPF	94	Hello Packet
19	42.099157	192.168.67.7	224.0.0.5	OSPF	94	Hello Packet
23	51.760053	192.168.67.6	224.0.0.5	OSPF	94	Hello Packet
24	52.100346	192.168.67.7	224.0.0.5	OSPF	94	Hello Packet
28	61.774385	192.168.67.6	224.0.0.5	OSPF	94	Hello Packet
29	62.095141	192.168.67.7	224.0.0.5	OSPF	94	Hello Packet
32	71.107009	192.168.67.6	224.0.0.5	OSPF	98	LS Update
33	71.114598	192.168.67.7	224.0.0.5	OSPF	98	LS Update
34	71.137269	192.168.67.6	224.0.0.5	OSPF	134	LS Update
35	71.156196	192.168.67.7	224.0.0.5	OSPF	134	LS Update
36	71.777482	192.168.67.6	224.0.0.5	OSPF	94	Hello Packet
37	72.081610	192.168.67.7	224.0.0.5	OSPF	94	Hello Packet
39	73.635469	192.168.67.7	224.0.0.5	OSPF	118	LS Acknowledge
40	73.642429	192.168.67.6	224.0.0.5	OSPF	78	LS Acknowledge
42	81.773420	192.168.67.6	224.0.0.5	OSPF	94	Hello Packet
43	82.083865	192.168.67.7	224.0.0.5	OSPF	94	Hello Packet
46	91.759909	192.168.67.6	224.0.0.5	OSPF	94	Hello Packet
47	92.082730	192.168.67.7	224.0.0.5	OSPF	94	Hello Packet
50	101.754020	192.168.67.6	224.0.0.5	OSPF	94	Hello Packet
51	102.104466	192.168.67.7	224.0.0.5	OSPF	94	Hello Packet
55	111.757465	192.168.67.6	224.0.0.5	OSPF	94	Hello Packet
56	112.079377	192.168.67.7	224.0.0.5	OSPF	94	Hello Packet
59	118.020278	192.168.67.7	224.0.0.5	OSPF	90	LS Update
61	120.512591	192.168.67.6	224.0.0.5	OSPF	78	LS Acknowledge
62	121.749877	192.168.67.6	224.0.0.5	OSPF	94	Hello Packet
63	122.106058	192.168.67.7	224.0.0.5	OSPF	94	Hello Packet

Рисунок 4 – Обмен сообщениями OSPF при восстановлении соседства маршрутизаторов

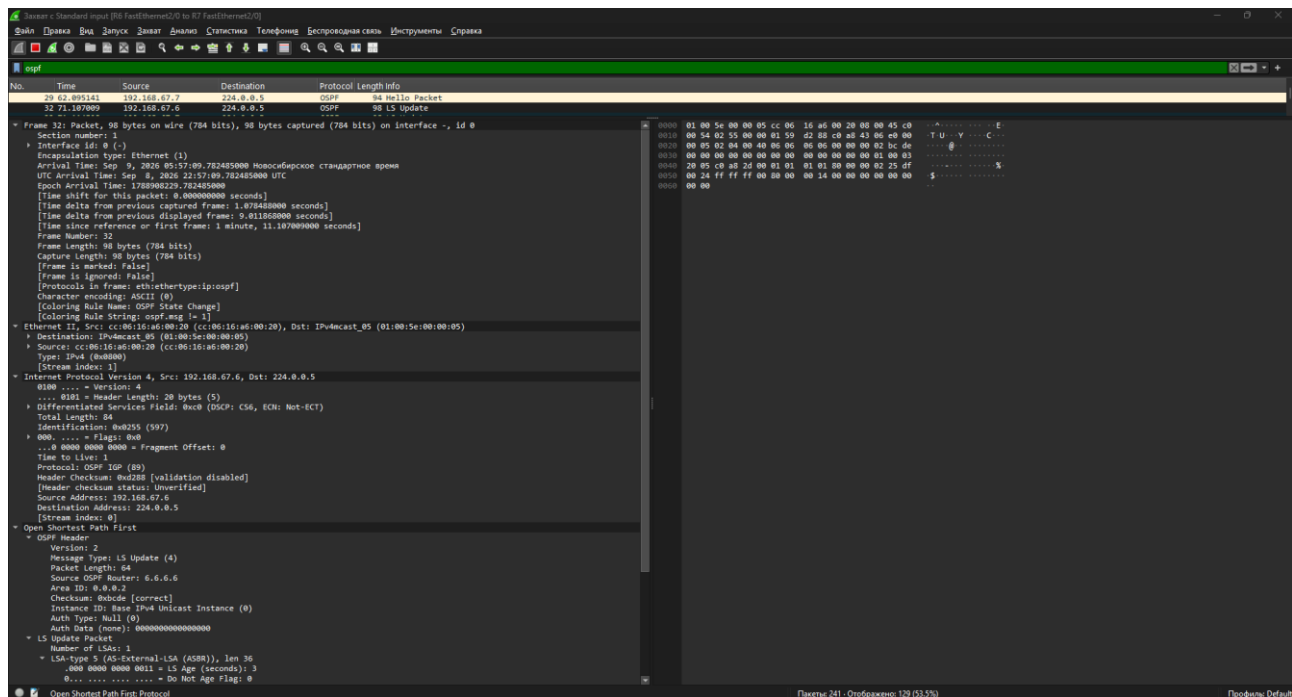


Рисунок 5 – Содержимое сообщения OSPF Link State Update

ЗАДАНИЕ 7

7) Сохранить в отдельные файлы с префиксом `rt_` и именем маршрутизатора таблицы маршрутизации всех маршрутизаторов.

R1#enable

R1#terminal length 0

R1#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C 192.168.123.0/24 is directly connected, FastEthernet1/0

R 192.168.45.0/24 [120/1] via 192.168.51.5, 00:00:11, FastEthernet0/0

O IA 192.168.26.0/24 [110/2] via 192.168.123.2, 00:07:39, FastEthernet1/0

O IA 192.168.38.0/24 [110/2] via 192.168.123.3, 00:43:16, FastEthernet1/0

O IA 192.168.4.0/24 [110/3] via 192.168.123.3, 00:07:39, FastEthernet1/0

O IA 192.168.5.0/24 [110/3] via 192.168.123.3, 00:43:16, FastEthernet1/0

O IA 192.168.67.0/24 [110/3] via 192.168.123.3, 00:07:49, FastEthernet1/0

[110/3] via 192.168.123.2, 00:07:51, FastEthernet1/0

O IA 192.168.37.0/24 [110/2] via 192.168.123.3, 00:07:41, FastEthernet1/0

C 192.168.51.0/24 is directly connected, FastEthernet0/0

R 192.168.1.0/24 [120/2] via 192.168.51.5, 00:00:12, FastEthernet0/0

R 192.168.2.0/24 [120/1] via 192.168.51.5, 00:00:12, FastEthernet0/0

O IA 192.168.3.0/24 [110/3] via 192.168.123.2, 00:07:40, FastEthernet1/0

R1#

R2#enable

R2#terminal length 0

R2#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C 192.168.123.0/24 is directly connected, FastEthernet0/0

O E2 192.168.45.0/24 [110/20] via 192.168.123.1, 00:08:54, FastEthernet0/0

C 192.168.26.0/24 is directly connected, FastEthernet1/0

O IA 192.168.38.0/24 [110/2] via 192.168.123.3, 00:08:54, FastEthernet0/0

O 192.168.4.0/24 [110/3] via 192.168.26.6, 00:08:54, FastEthernet1/0

O IA 192.168.5.0/24 [110/3] via 192.168.123.3, 00:08:54, FastEthernet0/0

O 192.168.67.0/24 [110/2] via 192.168.26.6, 00:08:54, FastEthernet1/0

O 192.168.37.0/24 [110/3] via 192.168.26.6, 00:08:55, FastEthernet1/0

O E2 192.168.51.0/24 [110/20] via 192.168.123.1, 00:08:55, FastEthernet0/0

O E2 192.168.1.0/24 [110/20] via 192.168.123.1, 00:08:55, FastEthernet0/0

O E2 192.168.2.0/24 [110/20] via 192.168.123.1, 00:08:55, FastEthernet0/0

O 192.168.3.0/24 [110/2] via 192.168.26.6, 00:08:55, FastEthernet1/0

R3#enable

R3#terminal length 0

R3#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C 192.168.123.0/24 is directly connected, FastEthernet0/0
O E2 192.168.45.0/24 [110/20] via 192.168.123.1, 00:10:09, FastEthernet0/0
O 192.168.26.0/24 [110/3] via 192.168.37.7, 00:10:09, FastEthernet1/0
C 192.168.38.0/24 is directly connected, FastEthernet2/0
O 192.168.4.0/24 [110/2] via 192.168.37.7, 00:10:09, FastEthernet1/0
O 192.168.5.0/24 [110/2] via 192.168.38.8, 00:57:56, FastEthernet2/0
O 192.168.67.0/24 [110/2] via 192.168.37.7, 00:10:09, FastEthernet1/0
C 192.168.37.0/24 is directly connected, FastEthernet1/0
O E2 192.168.51.0/24 [110/20] via 192.168.123.1, 00:10:10, FastEthernet0/0
O E2 192.168.1.0/24 [110/20] via 192.168.123.1, 00:10:10, FastEthernet0/0
O E2 192.168.2.0/24 [110/20] via 192.168.123.1, 00:10:10, FastEthernet0/0
O 192.168.3.0/24 [110/3] via 192.168.37.7, 00:10:10, FastEthernet1/0
R3#

R4#enable

R4#terminal length 0

R4#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

```
R 192.168.123.0/24 [120/3] via 192.168.45.5, 00:00:09, FastEthernet1/0
C 192.168.45.0/24 is directly connected, FastEthernet1/0
R 192.168.26.0/24 [120/3] via 192.168.45.5, 00:00:09, FastEthernet1/0
R 192.168.38.0/24 [120/3] via 192.168.45.5, 00:00:09, FastEthernet1/0
R 192.168.4.0/24 [120/3] via 192.168.45.5, 00:00:09, FastEthernet1/0
R 192.168.5.0/24 [120/3] via 192.168.45.5, 00:00:09, FastEthernet1/0
R 192.168.67.0/24 [120/3] via 192.168.45.5, 00:00:09, FastEthernet1/0
R 192.168.37.0/24 [120/3] via 192.168.45.5, 00:00:11, FastEthernet1/0
R 192.168.51.0/24 [120/1] via 192.168.45.5, 00:00:11, FastEthernet1/0
C 192.168.1.0/24 is directly connected, FastEthernet0/0
R 192.168.2.0/24 [120/1] via 192.168.45.5, 00:00:11, FastEthernet1/0
R 192.168.3.0/24 [120/3] via 192.168.45.5, 00:00:11, FastEthernet1/0
R4#
```

R5#enable

R5#terminal length 0

R5#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

```

R 192.168.123.0/24 [120/2] via 192.168.51.1, 00:00:04, FastEthernet2/0
C 192.168.45.0/24 is directly connected, FastEthernet1/0
R 192.168.26.0/24 [120/2] via 192.168.51.1, 00:00:04, FastEthernet2/0
R 192.168.38.0/24 [120/2] via 192.168.51.1, 00:00:04, FastEthernet2/0
R 192.168.4.0/24 [120/2] via 192.168.51.1, 00:00:04, FastEthernet2/0
R 192.168.5.0/24 [120/2] via 192.168.51.1, 00:00:04, FastEthernet2/0
R 192.168.67.0/24 [120/2] via 192.168.51.1, 00:00:04, FastEthernet2/0
R 192.168.37.0/24 [120/2] via 192.168.51.1, 00:00:07, FastEthernet2/0
C 192.168.51.0/24 is directly connected, FastEthernet2/0
R 192.168.1.0/24 [120/1] via 192.168.45.4, 00:00:21, FastEthernet1/0
C 192.168.2.0/24 is directly connected, FastEthernet0/0
R 192.168.3.0/24 [120/2] via 192.168.51.1, 00:00:07, FastEthernet2/0
R5#

```

R6#enable

R6#terminal length 0

R6#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

```

O IA 192.168.123.0/24 [110/2] via 192.168.26.2, 00:01:33, FastEthernet0/0
O E2 192.168.45.0/24 [110/20] via 192.168.26.2, 00:01:33, FastEthernet0/0
C 192.168.26.0/24 is directly connected, FastEthernet0/0
O IA 192.168.38.0/24 [110/3] via 192.168.67.7, 00:01:33, FastEthernet2/0

```

[110/3] via 192.168.26.2, 00:01:33, FastEthernet0/0
192.168.4.0/24 [11

R7#enable

R7#

*Mar 1 02:41:55.703: %OSPF-5-ADJCHG: Process 1, Nbr 6.6.6.6 on
FastEthernet2/0 from LOADING to FULL, Loading Done

R7#terminal length 0

R7#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

O IA 192.168.123.0/24 [110/2] via 192.168.37.3, 00:00:18, FastEthernet0/0

O E2 192.168.45.0/24 [110/20] via 192.168.37.3, 00:00:18, FastEthernet0/0

O 192.168.26.0/24 [110/2] via 192.168.67.6, 00:00:18, FastEthernet2/0

O IA 192.168.38.0/24 [110/2] via 192.168.37.3, 00:00:18, FastEthernet0/0

C 192.168.4.0/24 is directly connected, FastEthernet1/0

O IA 192.168.5.0/24 [110/3] via 192.168.37.3, 00:00:18, FastEthernet0/0

C 192.168.67.0/24 is directly connected, FastEthernet2/0

C 192.168.37.0/24 is directly connected, FastEthernet0/0

O E2 192.168.51.0/24 [110/20] via 192.168.37.3, 00:00:19, FastEthernet0/0

O E2 192.168.1.0/24 [110/20] via 192.168.37.3, 00:00:19, FastEthernet0/0

O E2 192.168.2.0/24 [110/20] via 192.168.37.3, 00:00:19, FastEthernet0/0

O 192.168.3.0/24 [110/2] via 192.168.67.6, 00:00:19, FastEthernet2/0
R7#

R6#enable

R6#terminal length 0

R6#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

O IA 192.168.123.0/24 [110/2] via 192.168.26.2, 00:13:18, FastEthernet0/0

O E2 192.168.45.0/24 [110/20] via 192.168.26.2, 00:13:18, FastEthernet0/0

C 192.168.26.0/24 is directly connected, FastEthernet0/0

O IA 192.168.38.0/24 [110/3] via 192.168.67.7, 00:13:18, FastEthernet2/0
[110/3] via 192.168.26.2, 00:13:18, FastEthernet0/0

O 192.168.4.0/24 [110/2] via 192.168.67.7, 00:13:18, FastEthernet2/0

O IA 192.168.5.0/24 [110/4] via 192.168.67.7, 00

*Mar 1 02:29:18.507: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on
FastEthernet0/0 from LOADING to FULL, Loading Done

*Mar 1 02:29:18.515: %OSPF-5-ADJCHG: Process 1, Nbr 7.7.7.7 on
FastEthernet2/0 from LOADING to FULL, Loading Done:13:18, FastEthernet2/0
[110/4] via 192.168.26.2, 00:13:23, FastEthernet0/0

C 192.168.67.0/24 is directly connected, FastEthernet2/0

O 192.168.37.0/24 [110/2] via 192.168.67.7, 00:13:23, FastEthernet2/0

O E2 192.168.51.0/24 [110/20] via 192.168.26.2, 00:13:23, FastEthernet0/0

O E2 192.168.1.0/24 [110/20] via 192.168.26.2, 00:13:23, FastEthernet0/0
O E2 192.168.2.0/24 [110/20] via 192.168.26.2, 00:13:23, FastEthernet0/0
C 192.168.3.0/24 is directly connected, FastEthernet1/0
R6#

R8#enable

R8#terminal length 0

R8#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is 192.168.38.3 to network 0.0.0.0

C 192.168.38.0/24 is directly connected, FastEthernet0/0

C 192.168.5.0/24 is directly connected, FastEthernet1/0

O*IA 0.0.0.0/0 [110/2] via 192.168.38.3, 01:13:20, FastEthernet0/0

R8#

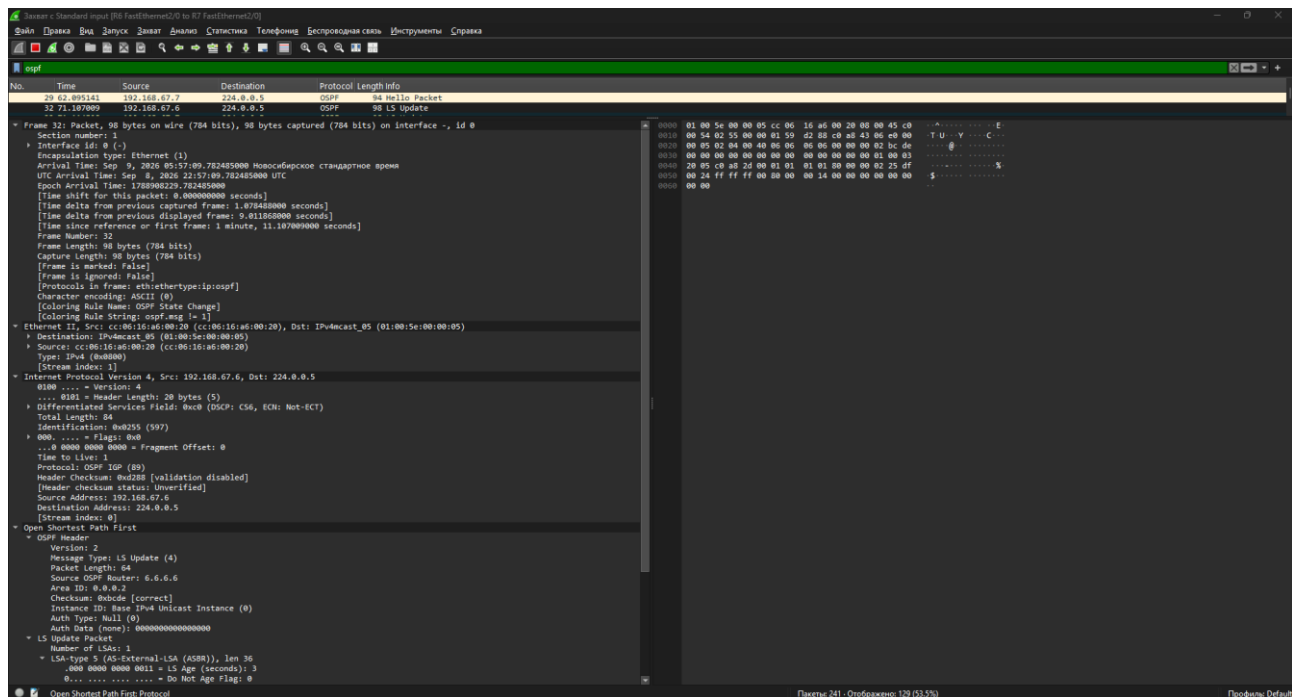


Рисунок 6 – Файлы с сохранёнными таблицами маршрутизации маршрутизаторов R1–R8

ЗАДАНИЕ 8

8) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств.

```
vIOS-L2-01>enable
```

```
vIOS-L2-01#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
vIOS-L2-01(config)#
```

```
*Sep  8 22:03:12.574: %CDP-4-DUPLEX_MISMATCH: duplex mismatch  
discovered on GigabitEthernet0/2 (not full duplex), with R3 FastEthernet0/0 (full  
duplex).
```

```
vIOS-L2-01(config)#interface vlan 1
```

```
vIOS-L2-01(config-if)#
```

```
*Sep   8 22:03:15.759: %LINEPROTO-5-UPDOWN: Line protocol on  
Interface Vlan1, changed state to down
```

```
vIOS-L2-01(config-if)#ip address 192.168.123.254 255.255.255.0
```

```
vIOS-L2-01(config-if)# no shutdown
```

```
vIOS-L2-01(config-if)#
```

```
*Sep  8 22:03:21.551: %CDP-4-DUPLEX_MISMATCH: duplex mismatch  
discovered on GigabitEthernet0/0 (not full duplex), with R1 FastEthernet1/0 (full  
duplex).
```

```
vIOS-L2-01(config-if)#exit
```

```
vIOS-L2-01(config)#
```

```
*Sep  8 22:03:23.658: %LINK-3-UPDOWN: Interface Vlan1, changed state to  
up
```

```
vIOS-L2-01(config)#
```

```
*Sep   8 22:03:24.658: %LINEPROTO-5-UPDOWN: Line protocol on  
Interface Vlan1, changed state to up
```

```
vIOS-L2-01(config)#ip default-gateway 192.168.123.1
```

```
vIOS-L2-01(config)#
```

*Sep 8 22:03:25.516: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on GigabitEthernet0/1 (not full duplex), with R2 FastEthernet0/0 (full duplex).

vIOS-L2-01(config)#end

vIOS-L2-01#

*Sep 8 22:03:30.738: %SYS-5-CONFIG_I: Configured from console by console

vIOS-L2-01#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 86CE9Ez 8679B9z 8574D3z 85914Ez - Process "Exec", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

Compressed configuration from 4936 bytes to 1902 bytes[OK]

*Sep 8 22:03:38.417: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than (2000)msecs (26/17),process = Async write process.

*Sep 8 22:03:41.441: %SYS-3-CPUHOG: Task is running for (2000)msecs, more than (2000)msecs (9/0),process = Async write process.

*Sep 8 22:03:45.017: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than (2000)msecs (6/0),process = Async write process.

*Sep 8 22:03:47.841: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than (2000)msecs (16/6),process = Async write process.

*Sep 8 22:03:50.557: %SYS-3-CPUHOG: Task is running for (2000)msecs, more than (2000)msecs (11/0),process = Async write process.

*Sep 8 22:03:51.107: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on GigabitEthernet0/2 (not full duplex), with R3 FastEthernet0/0 (full duplex).

*Sep 8 22:03:53.189: %SYS-3-CPUHOG: Task is running for (1999)msecs, more than (2000)msecs (22/12),process = Async write process.

*Sep 8 22:03:56.865: %SYS-3-CPUHOG: Task is running for (2000)msecs, more than (2000)msecs (7/0),process = Exec.

*Sep 8 22:04:00.053: %SYS-3-CPUHOG: Task is running for (2000)msecs, more than (2000)msecs (11/0),process = Async write process.

*Sep 8 22:04:00.668: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...

*Sep 8 22:04:01.244: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on GigabitEthernet0/0 (not full duplex), with R1 FastEthernet1/0 (full duplex).

*Sep 8 22:04:02.790: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.

vIOS-L2-01#show ip interface brief

*Sep 8 22:04:05.027: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on GigabitEthernet0/1 (not full duplex), with R2 FastEthernet0/0 (full duplex).

vIOS-L2-01#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up
GigabitEthernet0/1	unassigned	YES	unset	up	up
GigabitEthernet0/2	unassigned	YES	unset	up	up
Vlan1	192.168.123.254	YES	manual	up	up

vIOS-L2-01#ping 192.168.123.1

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.123.1, timeout is 2 seconds:

.!!!!

Success rate is 80 percent (4/5), round-trip min/avg/max = 6/9/11 ms

vIOS-L2-01#ping 192.168.123.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.123.2, timeout is 2 seconds:

.!!!!

Success rate is 80 percent (4/5), round-trip min/avg/max = 2/10/19 ms

vIOS-L2-01#ping 192.168.123.3

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.123.3, timeout is 2 seconds:

..!!!!

Success rate is 80 percent (4/5), round-trip min/avg/max = 4/11/23 ms

vIOS-L2-01#

*Sep 8 22:04:25.472: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on GigabitEthernet0/2 (not full duplex), with R3 FastEthernet0/0 (full duplex).

vIOS-L2-01#

vIOS-L2-01#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	1	auto	auto	unknown
Gi0/1		connected	1	auto	auto	unknown
Gi0/2		connected	1	auto	auto	unknown

vIOS-L2-01#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up
GigabitEthernet0/1	unassigned	YES	unset	up	up
GigabitEthernet0/2	unassigned	YES	unset	up	up
Vlan1	192.168.123.254	YES	manual	up	up

vIOS-L2-01#

*Sep 8 22:08:03.252: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on GigabitEthernet0/0 (not full duplex), with R1 FastEthernet1/0 (full duplex).

vIOS-L2-01#ping 192.168.123.1

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.123.1, timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 2/4/11 ms

vIOS-L2-01#

*Sep 8 22:08:07.281: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on GigabitEthernet0/1 (not full duplex), with R2 FastEthernet0/0 (full duplex).

vIOS-L2-01#

vIOS-L2-01#enable

vIOS-L2-01#

*Sep 8 22:09:02.548: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on GigabitEthernet0/2 (not full duplex), with R3 FastEthernet0/0 (full duplex).

vIOS-L2-01#terminal length 0

vIOS-L2-01#show running-config

Building configuration...

Current configuration : 4936 bytes

!

! Last configuration change at 22:07:16 UTC Tue Sep 8 2026

!

version 15.0

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

service compress-config

!

hostname vIOS-L2-01

!

boot-start-marker

boot-end-marker

!

!

```
!  
no aaa new-model  
!  
!  
!  
!  
!  
!  
vtp domain CISCO-vIOS  
vtp mode transparent  
!  
!  
!  
ip cef  
no ipv6 cef  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
vlan internal allocation policy ascending  
!  
vlan 100  
    name VLAN100  
!  
vlan 200,300  
!  
!  
!  
!  
!
```

```
!  
!  
!  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
    media-type rj45  
    negotiation auto  
!  
interface GigabitEthernet0/1  
    media-type rj45  
    negotiation auto  
!  
interface GigabitEthernet0/2  
    media-type rj45  
    negotiation auto  
!  
interface Vlan1  
    ip address 192.168.123.254 255.255.255.0  
!  
    ip default-gateway 192.168.123.1  
    ip forward-protocol nd  
!  
    no ip http server  
    no ip http secure-server  
!  
!
```

!
!
!
!
control-plane
!
banner exec ^C

* IOSv - Cisco Systems Confidential *

* *

* This software is provided as is without warranty for internal *

* development and testing purposes only under the terms of the Cisco *

* Early Field Trial agreement. Under no circumstances may this software *

* be used for production purposes or deployed in a production *

* environment. *

* *

* By using the software, you agree to abide by the terms and conditions *

* of the Cisco Early Field Trial Agreement as well as the terms and *

* conditions of the Cisco End User License Agreement at *

* <http://www.cisco.com/go/eula> *

* *

* Unauthorized use or distribution of this software is expressly *

* Prohibited. *

*****^C

banner incoming ^C

* IOSv - Cisco Systems Confidential *

```

*
*
* This software is provided as is without warranty for internal
* development and testing purposes only under the terms of the Cisco
* Early Field Trial agreement. Under no circumstances may this software
* be used for production purposes or deployed in a production
* environment.
*
*
* By using the software, you agree to abide by the terms and conditions
* of the Cisco Early Field Trial Agreement as well as the terms and
* conditions of the Cisco End User License Agreement at
* http://www.cisco.com/go/eula
*
*
* Unauthorized use or distribution of this software is expressly
* Prohibited.

```

```

*****

```

```

*****^C

```

```

banner login ^C

```

```

*****

```

```

*****

```

```

* IOSv - Cisco Systems Confidential
*
*
* This software is provided as is without warranty for internal
* development and testing purposes only under the terms of the Cisco
* Early Field Trial agreement. Under no circumstances may this software
* be used for production purposes or deployed in a production
* environment.
*
*
* By using the software, you agree to abide by the terms and conditions
* of the Cisco Early Field Trial Agreement as well as the terms and
* conditions of the Cisco End User License Agreement at

```

```

* http://www.cisco.com/go/eula
*
*
* Unauthorized use or distribution of this software is expressly
* Prohibited.
*
*****
*****^C
!
line con 0
 logging synchronous
line aux 0
line vty 0 4
 logging synchronous
 login
line vty 5 15
 logging synchronous
 login
!
!
end

vIOS-L2-01#
*Sep  8 22:09:09.169: %CDP-4-DUPLEX_MISMATCH: duplex mismatch
discovered on GigabitEthernet0/0 (not full duplex), with R1 FastEthernet1/0 (full
duplex).

vIOS-L2-01#
*Sep  8 22:09:12.565: %CDP-4-DUPLEX_MISMATCH: duplex mismatch
discovered on GigabitEthernet0/1 (not full duplex), with R2 FastEthernet0/0 (full
duplex).

vIOS-L2-01#

```

R1#terminal length 0

R1#show running-config

Building configuration...

Current configuration : 1023 bytes

!

version 12.4

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

!

hostname R1

!

boot-start-marker

boot-end-marker

!

!

no aaa new-model

memory-size iomem 5

no ip icmp rate-limit unreachable

!

!

ip cef

no ip domain lookup

!

!

!

!

!

!

```
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
interface FastEthernet0/0  
  ip address 192.168.51.1 255.255.255.0  
  duplex auto  
  speed auto  
!  
interface FastEthernet1/0  
  ip address 192.168.123.1 255.255.255.0  
  duplex auto  
  speed auto  
!  
!  
router ospf 1  
  router-id 1.1.1.1  
  log-adjacency-changes  
  redistribute rip subnets  
  network 192.168.123.0 0.0.0.255 area 0
```

```
!  
router rip  
  version 2  
  redistribute ospf 1 metric 2  
  network 192.168.51.0  
  no auto-summary  
!  
no ip http server  
ip forward-protocol nd  
!  
!  
!  
no cdp log mismatch duplex  
!  
!  
!  
control-plane  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
line con 0  
  exec-timeout 0 0  
  privilege level 15  
  logging synchronous
```

```
line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line vty 0 4
  login
  !
  !
```

End

```
R2#terminal length 0
R2#show running-config
Building configuration...

Current configuration : 943 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R2
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
memory-size iomem 5
no ip icmp rate-limit unreachable
```

```
!  
!  
ip cef  
no ip domain lookup  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
interface FastEthernet0/0  
ip address 192.168.123.2 255.255.255.0  
duplex auto  
speed auto  
!  
interface FastEthernet1/0
```

```
ip address 192.168.26.2 255.255.255.0
duplex auto
speed auto
!
!
router ospf 1
router-id 2.2.2.2
log-adjacency-changes
network 192.168.26.0 0.0.0.255 area 2
network 192.168.123.0 0.0.0.255 area 0
!
no ip http server
ip forward-protocol nd
!
!
!
no cdp log mismatch duplex
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
```

```
line con 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line vty 0 4
  login
  !
  !
```

End

```
R6#terminal length 0
R6#show running-config
Building configuration...

Current configuration : 1105 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R6
!
boot-start-marker
boot-end-marker
!
!
```

[illegible]

```
speed auto
!
interface FastEthernet1/0
ip address 192.168.3.1 255.255.255.0
duplex auto
speed auto
!
interface FastEthernet2/0
ip address 192.168.67.6 255.255.255.0
duplex auto
speed auto
!
!
router ospf 1
router-id 6.6.6.6
log-adjacency-changes
passive-interface FastEthernet1/0
network 192.168.3.0 0.0.0.255 area 2
network 192.168.26.0 0.0.0.255 area 2
network 192.168.67.0 0.0.0.255 area 2
!
no ip http server
ip forward-protocol nd
!
!
!
no cdp log mismatch duplex
!
!
!
```

```
control-plane
!
!
!
!
!
!
!
!
!
line con 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line vty 0 4
  login
  !
  !
end
```

R6#

R7#enable

R7#

*Mar 1 02:41:55.703: %OSPF-5-ADJCHG: Process 1, Nbr 6.6.6.6 on
FastEthernet2/0 from LOADING to FULL, Loading Done

R7#terminal length 0

R7#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

O IA 192.168.123.0/24 [110/2] via 192.168.37.3, 00:00:18, FastEthernet0/0

O E2 192.168.45.0/24 [110/20] via 192.168.37.3, 00:00:18, FastEthernet0/0

O 192.168.26.0/24 [110/2] via 192.168.67.6, 00:00:18, FastEthernet2/0

O IA 192.168.38.0/24 [110/2] via 192.168.37.3, 00:00:18, FastEthernet0/0

C 192.168.4.0/24 is directly connected, FastEthernet1/0

O IA 192.168.5.0/24 [110/3] via 192.168.37.3, 00:00:18, FastEthernet0/0

C 192.168.67.0/24 is directly connected, FastEthernet2/0

C 192.168.37.0/24 is directly connected, FastEthernet0/0

O E2 192.168.51.0/24 [110/20] via 192.168.37.3, 00:00:19, FastEthernet0/0

O E2 192.168.1.0/24 [110/20] via 192.168.37.3, 00:00:19, FastEthernet0/0

O E2 192.168.2.0/24 [110/20] via 192.168.37.3, 00:00:19, FastEthernet0/0

O 192.168.3.0/24 [110/2] via 192.168.67.6, 00:00:19, FastEthernet2/0

R7#

*Mar 1 02:43:50.395: %OSPF-5-ADJCHG: Process 1, Nbr 6.6.6.6 on
FastEthernet2/0 from FULL to DOWN, Neighbor Down: Dead timer expired

R7#

*Mar 1 02:55:35.443: %OSPF-5-ADJCHG: Process 1, Nbr 6.6.6.6 on
FastEthernet2/0 from LOADING to FULL, Loading Done

R7#terminal length 0

R7#show running-config

Building configuration...

Current configuration : 1105 bytes

!

version 12.4

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

!

hostname R7

!

boot-start-marker

boot-end-marker

!

!

no aaa new-model

memory-size iomem 5

no ip icmp rate-limit unreachable

!

!

ip cef

no ip domain lookup

!

!

!

!

!

!

!

```
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
interface FastEthernet0/0  
  ip address 192.168.37.7 255.255.255.0  
  duplex auto  
  speed auto  
!  
interface FastEthernet1/0  
  ip address 192.168.4.1 255.255.255.0  
  duplex auto  
  speed auto  
!  
interface FastEthernet2/0  
  ip address 192.168.67.7 255.255.255.0  
  duplex auto  
  speed auto  
!  
!  
router ospf 1
```

```
router-id 7.7.7.7
log-adjacency-changes
passive-interface FastEthernet1/0
network 192.168.4.0 0.0.0.255 area 2
network 192.168.37.0 0.0.0.255 area 2
network 192.168.67.0 0.0.0.255 area 2
!
no ip http server
ip forward-protocol nd
!
!
!
no cdp log mismatch duplex
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
```

```
line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line vty 0 4
  login
  !
  !
end
```

```
R8#terminal length 0
```

```
R8#show running-config
```

```
Building configuration...
```

```
Current configuration : 987 bytes
```

```
!
```

```
version 12.4
```

```
service timestamps debug datetime msec
```

```
service timestamps log datetime msec
```

```
no service password-encryption
```

```
!
```

```
hostname R8
```

```
!
```

```
boot-start-marker
```

```
boot-end-marker
```

```
!
```

```
!
```

```
no aaa new-model
```

```
memory-size iomem 5
```

```
no ip icmp rate-limit unreachable
```

```
!
```

```
!  
ip cef  
no ip domain lookup  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
interface FastEthernet0/0  
ip address 192.168.38.8 255.255.255.0  
duplex auto  
speed auto  
!  
interface FastEthernet1/0  
ip address 192.168.5.1 255.255.255.0
```

```
duplex auto
speed auto
!
!
router ospf 1
router-id 8.8.8.8
log-adjacency-changes
area 1 stub
passive-interface FastEthernet1/0
network 192.168.5.0 0.0.0.255 area 1
network 192.168.38.0 0.0.0.255 area 1
!
no ip http server
ip forward-protocol nd
!
!
!
no cdp log mismatch duplex
!
!
!
!
!
!
!
!
!
!
```

```
!  
line con 0  
  exec-timeout 0 0  
  privilege level 15  
  logging synchronous  
line aux 0  
  exec-timeout 0 0  
  privilege level 15  
  logging synchronous  
line vty 0 4  
  login  
!  
!  
End
```

```
PC1> show ip
```

```
NAME      : PC1[1]  
IP/MASK    : 192.168.1.10/24  
GATEWAY    : 192.168.1.1  
DNS        :  
MAC        : 00:50:79:66:68:00  
LPORT     : 27304  
RHOST:PORT : 127.0.0.1:27305  
MTU        : 1500
```

```
PC2> show ip
```

```
NAME      : PC2[1]  
IP/MASK    : 192.168.2.10/24  
GATEWAY    : 192.168.2.1  
DNS        :
```

MAC : 00:50:79:66:68:01
LPORT : 27306
RHOST:PORT : 127.0.0.1:27307
MTU : 1500

PC4> show ip

NAME : PC4[1]
IP/MASK : 192.168.4.10/24
GATEWAY : 192.168.4.1
DNS :
MAC : 00:50:79:66:68:03
LPORT : 27310
RHOST:PORT : 127.0.0.1:27311
MTU : 1500

PC5> show ip

NAME : PC5[1]
IP/MASK : 192.168.5.10/24
GATEWAY : 192.168.5.1
DNS :
MAC : 00:50:79:66:68:04
LPORT : 27312
RHOST:PORT : 127.0.0.1:27313
MTU : 1500

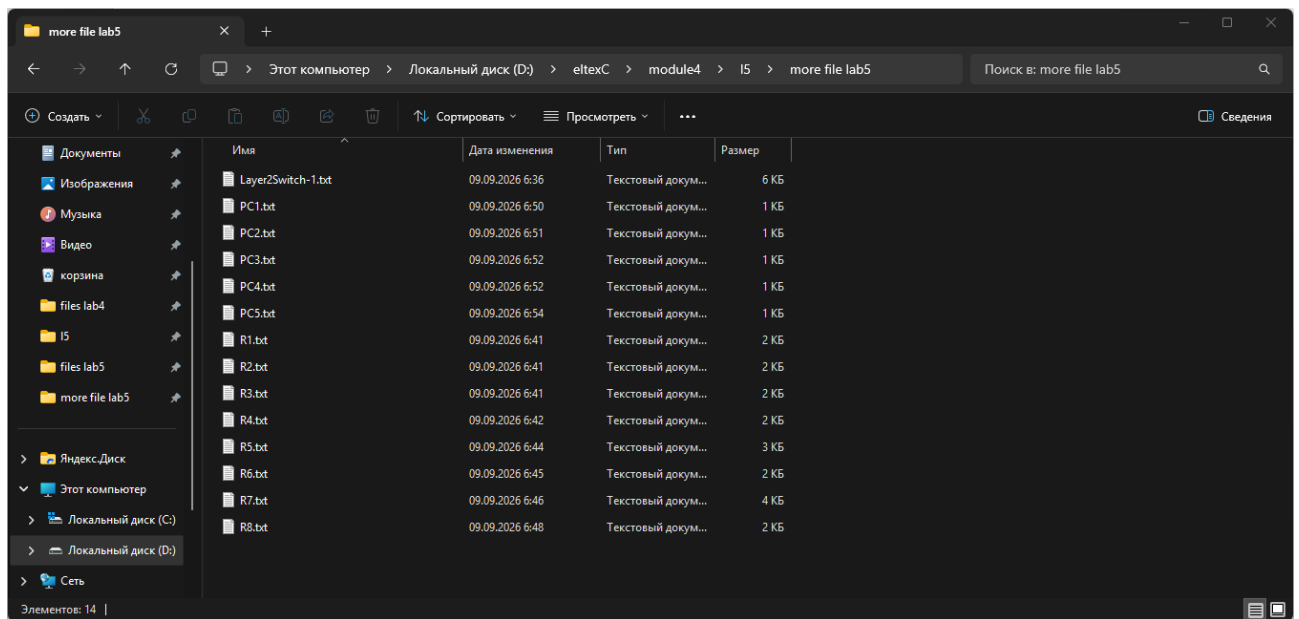


Рисунок 7 – Файлы с сохранёнными конфигурациями устройств сети